

Supplement: Return forecasts and execution timing

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This supplement documents the exploratory extension, complete comparisons, proofs and related literature. Quantities called gains are paired reductions in a hypothetical execution objective, not realised trading profits.

1 Chronology and reproducible design

The initial study used Bitcoin in June, July and August 2026 for training, calibration and evaluation. Its results and peer-review reports existed before this extension was designed. The extension protocol was recorded before computing the expanded outcomes, but after inspecting that original result. It is therefore retrospective and exploratory, not preregistered or an independent confirmation. Bitcoin’s August window overlaps the original analysis. All twelve asset-month windows, four forecast constructions and six cost scenarios were fixed in the extension protocol. No additional profile search or outcome-driven sample expansion was performed.

Public Binance monthly spot archives contain one-minute BTCUSDT and ETHUSDT bars for January–August 2026 [Binance, 2026]. All sixteen archive hashes matched their published checksums. Validation covered all 699,840 rows: twelve columns, finite entries, exact calendar boundaries, microsecond timestamps spaced by one minute, no duplicates or gaps, close-time consistency, positive prices, OHLC bounds, nonnegative volumes and integer trade counts, and taker volumes bounded by corresponding totals. Positional indexing occurred only after validation. The manifest records each archive URL, hash, row count and individual checks. An initial parser interface error was corrected before outcomes were produced; its history is recorded separately.

Let i index the decision minute, at minute offsets 5, 35, $\dots$, 1415 each UTC day. If V_k and V_k^B are total and taker-buy base volume, the signal is

$$x_i = \frac{\sum_{k=i-5}^{i-1} (2V_k^B - V_k)}{\sum_{k=i-5}^{i-1} V_k},$$

with zero assigned for a zero denominator. These five bars are completed before the decision. Execution prices are opens at $i + 1, \dots, i + 15$, giving a one-minute initial delay. Returns in basis points use the fixed decision-minute open as denominator. There are 48 nonoverlapping execution episodes per day per asset. They may nevertheless be statistically dependent.

For each asset, fit fifteen regressions with intercept on the training month, one for each subsequent price horizon. If $\hat{\beta}_j$ is the slope and $\bar{x}_{\text{train}}$ the training signal mean, use $\mu_{ij} = \hat{\beta}_j(x_i - \bar{x}_{\text{train}})$. The unconditional fitted intercept is deliberately excluded. Training, calibration and evaluation use three consecutive months, starting with January, February and March. The window then advances one month at a time, ending with June, July and August. No evaluation observations enter that window’s fit or calibration. Rolling windows share observations and regimes; twelve rows do not represent twelve independent experiments.

For each profile solve the nonnegative, unit-volume quadratic programme in the main text. The baseline minimises the same cost without price information. An equality-constrained solution is used when feasible; otherwise the original projected-gradient simplex solver is used. Maximum observed KKT residual was 1.29×10^{-12} . The exact parallel-profile invariance is also a numerical control: absolute pooled mean residuals were below 1.5×10^{-14} basis points. Tables display their analytical zero. Code and protocol hashes were unchanged during the completed run.

2 Complete empirical comparisons

All forecast variants have identical final coordinates. Endpoint predictions, squared errors and directional-hit indicators coincide episode by episode, so aggregate RMSE, hit rate and R^2 coincide within each window. Endpoint R^2 is $1 - \sum_i (p_{iN} - \mu_{iN})^2 / \sum_i p_{iN}^2$, relative to a zero-return prediction. Hit rate compares forecast and realised terminal signs. The ramp is a restricted constructed profile; the mirror deliberately reverses temporal contrasts. Neither is represented as an independently trained competitor. The diagnostic holds terminal predictions fixed to expose omitted decision information.

Table 1: All primary evaluation windows ($\eta = 2, r = 1$). Gains are objective basis points; weights are fitted on the previous month. Months are in 2026.

Asset	Month	Hit rate	Terminal R^2	Learned	Ramp	w_P	w_C
BTC	03	51.9%	-0.00005	0.00184	0.00024	0.000	0.000
BTC	04	49.0%	-0.00452	-0.03025	-0.01944	0.000	0.000
BTC	05	52.6%	-0.00037	-0.00648	-0.00086	0.058	0.000
BTC	06	48.8%	-0.00022	0.00003	0.00168	0.000	0.000
BTC	07	50.0%	0.00007	-0.00035	0.00001	0.520	0.000
BTC	08	49.5%	-0.00127	0.00488	-0.00889	0.000	0.000
ETH	03	50.5%	-0.00046	-0.00535	-0.00224	0.000	0.000
ETH	04	47.9%	-0.00006	-0.02624	-0.00140	0.049	0.000
ETH	05	50.5%	0.00034	-0.00809	0.00434	0.000	0.000
ETH	06	48.4%	-0.00002	-0.00072	0.00273	0.620	0.000
ETH	07	51.1%	0.00189	0.00695	0.00692	0.824	0.000
ETH	08	50.7%	0.00070	0.01023	0.00508	0.674	0.000

The pooled comparison contains 17,664 episodes over 184 UTC dates. Equal episode weights imply modestly different calendar-month weights. Whole-day bootstrap sampling retains the two assets together on each sampled date, preserving their within-day co-movement. Each interval uses 2,000 draws, with policies and estimates held fixed; the evaluation seed is 20260920. This procedure does not account for arbitrary cross-day dependence, full retraining uncertainty or the choice of this study after the original analysis. No multiplicity adjustment is made. Intervals are conditional descriptive summaries.

The plug-in policy maximises the preceding month’s average quadratic gain on $[0, 1]$. The lower-percentile policy clips the fifth percentile of 2,000 day-bootstrap ratios to that interval, using calibration seed 20260919. Bootstrap samples aggregate coefficients before taking the ratio. The method does not average episode-level ratios. Identical zero gains for the lower-percentile policy

mean it equals the baseline in every window, not that future execution costs have zero uncertainty.

Table 2: Primary pooled exposure comparisons. Monetary gain removes the inventory penalty when scoring the same schedules; it is distinct from the optimisation objective. All entries are basis points.

Exposure policy	Objective gain	95% interval	Monetary gain
Learned	-0.00436	[-0.00829, -0.00047]	-0.00411
Half learned	-0.00017	[-0.00196, 0.00171]	-0.00007
Plug-in	0.00124	[-0.00035, 0.00285]	0.00125
Lower-percentile	0.00000	[0.00000, 0.00000]	0.00000

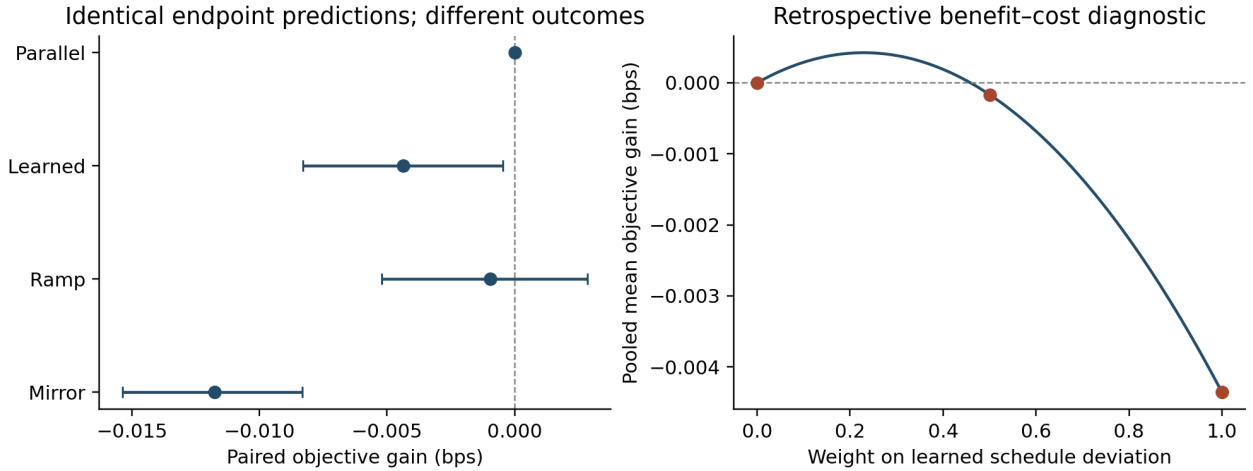


Figure 1: Matched-endpoint objective gains (left) and the retrospectively evaluated common-weight gain curve (right), under primary hypothetical costs. Orange markers indicate the reference exposure weights $w = 0$, $w = 1/2$ and $w = 1$; they are not fitted policy estimates. The right panel uses evaluation outcomes to describe the benefit–cost trade-off; its maximum is hindsight information, not an implementable fitted policy or an additional trading result.

Average learned-profile price alignment is 0.00369553 bps and curvature cost is 0.00805194 bps. The baseline first-order term vanishes to numerical precision. Average shifted quantity, measured as half the L^1 distance between schedules, is approximately 1.82% of the parent order. The learned full-exposure gain is positive in five of twelve windows, and negative when pooled. Thus neither endpoint accuracy nor positive gross alignment establishes a net benefit. The hypothesis of a useful new universally superior execution rule is not supported.

Full learned exposure has a negative pooled point estimate in all six scenarios. The smallest-impact, zero-inventory-penalty interval includes zero. The others do not under this conditional resampling design. This grid cannot substitute for estimating impact from transactions, and no scenario is chosen because its interval is more favourable. The full result file includes every policy, pairwise diagnostic contrast, decomposition, window weight and terminal diagnostic. The compressed episode file contains all primary episode comparisons.

A separate reproduction retains the original calibration and evaluation bootstrap seeds. Its maximum absolute discrepancy for the checked original ratios and full/half-exposure intervals was

Table 3: All six hypothetical cost specifications. Shared observations mean these are sensitivity checks, not independent discoveries. Full profile and exposure results for every scenario are retained in the machine-readable results.

η	r	Learned gain	95% interval	Plug-in gain
0.5	0	-0.01420	$[-0.02929, 0.00141]$	0.00533
0.5	1	-0.01562	$[-0.02908, -0.00193]$	0.00409
2	0	-0.00427	$[-0.00833, -0.00024]$	0.00133
2	1	-0.00436	$[-0.00829, -0.00047]$	0.00124
5	0	-0.00171	$[-0.00333, -0.00010]$	0.00053
5	1	-0.00172	$[-0.00333, -0.00014]$	0.00052

1.39×10^{-17} . New-bootstrap intervals need not equal original-bootstrap intervals. The new solver reproduced the original full/half mean gains within 6.1×10^{-14} . Original source files, data locks and results were preserved.

3 Proofs and economic interpretation

3.1 Forecast levels and endpoint insufficiency

For any fixed-volume feasible set $\mathcal{U} \subset \{u : \mathbf{1}^\top u = 1\}$,

$$K(u) - (\mu + a\mathbf{1})^\top u = K(u) - \mu^\top u - a.$$

The additive constant leaves the entire minimiser set unchanged. Likewise, for two feasible schedules u, v , $(u - v)^\top (p + a\mathbf{1}) = (u - v)^\top p$. This is price-level invariance, not invariance to a drift that changes prices differently across slots. Forecast-dependent fees, constraints or impact coefficients can invalidate the stated argument.

An endpoint μ_N leaves $N - 1$ temporal contrasts unspecified. The two-slot example in the article makes that omission explicit. For $K = \|u\|^2$ and $u = (x, 1 - x)$, differentiating gives $4x - 2 - (\mu_1 - \mu_2) = 0$, clipped to $[0, 1]$. With prices $(3, 2)$, flat, correct-shape and opposite-shape forecasts yield the reported 0, 0.125, -0.375 gains. These are synthetic dimensionless values. Additional structural restrictions can link the endpoint to all intermediate prices; the example does not deny their usefulness.

3.2 Benefit, baseline penalty and curvature

Write $H = \nabla^2 K = 2\eta NI + 2rL^\top L/N$, $q_0 = \mathbf{1} - Lu_0$ and $\delta = u_1 - u_0$. Direct expansion gives

$$\begin{aligned} A &= \delta^\top p, \\ B &= 2\eta Nu_0^\top \delta - \frac{2r}{N} q_0^\top L\delta, \\ C &= \eta N \|\delta\|^2 + \frac{r}{N} \|L\delta\|^2, \\ J_p(u_0) - J_p(u_0 + w\delta) &= w(A - B) - w^2 C. \end{aligned}$$

Convex baseline optimality implies $B \geq 0$ for every feasible direction to another schedule. On an interior simplex optimum, $\nabla K(u_0)$ is a multiple of $\mathbf{1}$, and $\mathbf{1}^\top \delta = 0$ gives $B = 0$. This argument concerns the same inventory-adjusted cost that defines the baseline. It need not give the same sign for monetary costs after dropping the inventory term.

Let $D_j = \sum_{k \leq j} \delta_k$ with $D_N = 0$. Since $\delta_j = D_j - D_{j-1}$, summation by parts gives $A = -\sum_{j < N} D_j(p_{j+1} - p_j)$. Earlier sales have positive D_j and benefit from subsequent declines. The price move before the first execution slot cancels because both schedules still hold the full order. This is an accounting identity, not evidence of causal market impact.

3.3 Historical fitting and its limits

For $C_i > 0$, set $z_i = (A_i - B_i)/(2C_i)$. Then

$$G_i(w) = C_i z_i^2 - C_i (w - z_i)^2.$$

A common gain-maximising weight therefore minimises a curvature-weighted squared loss and equals the clipped aggregate ratio in the article. It is not the mean of clipped episode ratios. With $\eta > 0$, $C_i = 0$ implies $\delta_i = 0$ and a zero gain contribution. The z_i are ex-post targets, unavailable at decision time; only historical observations can be used for policy fitting. This identity supplies no generalisation or coverage theorem.

Schedule mixing also differs from reoptimising scaled forecasts when constraints bind. Take the two-slot cost above and $\mu = (6, 2)$. The full solution is $(1, 0)$ and the baseline $(1/2, 1/2)$. At half exposure their mixture is $(3/4, 1/4)$. The endpoint-preserving half-scaled forecast $(4, 2)$ instead optimises at $(1, 0)$. Reoptimisation and mixing can agree within an unchanged active set, but not generally across active-set changes.

4 Technical corrections to the related preprint

The predecessor [Lai, 2026] considers an unrestricted OU-signal execution model. Its kernel requires correction before that model is used as context. Let $\tau = T - t$, $\kappa = \sqrt{r/\eta}$, and let $\phi \geq 0$ denote signal mean reversion. The regular solution of

$$B_t - [\kappa \coth(\kappa(T - t)) + \phi]B + 1 = 0 \tag{1}$$

is

$$B(t) = \int_t^T e^{-\phi(s-t)} \frac{\sinh(\kappa(T-s))}{\sinh(\kappa(T-t))} ds. \tag{2}$$

An integrating factor $e^{\phi\tau} \sinh(\kappa\tau)$ proves the expression. At $\kappa = 0$, the ratio is interpreted as $(T-s)/(T-t)$; near completion $B(T-\tau) = \tau/2 + O(\tau^2)$. The inverted ratio printed in the predecessor has a reciprocal endpoint singularity and a divergent improper integral. It is not a valid numerical comparator.

The side convention also matters: $+fx$ is the waiting-cost drift term for acquisition, whereas liquidation uses $-fx$. A deterministic linear permanent-impact term can be removed from fixed-endpoint comparisons when it is a schedule-independent constant; this simplification does not extend automatically to transient or nonlinear impact. The unrestricted model allows signed trades and inventory crossings, unlike the empirical simplex problem.

A nonnegative-control Hamiltonian yields $\max(0, V_x/(2\eta))$ using the *constrained* value function. Substituting the unrestricted quadratic value function does not solve the constrained problem. A

deterministic acquisition example makes the distinction concrete. Set $\eta = Q = T = 1$, $r = 0$, and $f(t) = 16e^{-0.1t}$. The buy-only optimum has

$$v^*(t) = 80(e^{-0.1t} - e^{-0.1z})\mathbf{1}_{\{t \leq z\}}, \quad \int_0^z v^*(t) dt = 1. \quad (3)$$

Convex first-order conditions establish this optimum. It completes at approximately 0.508529 with cost 5.299888. Corrected unrestricted feedback, clipped and stopped at zero inventory, completes at 0.260671 with cost 5.811578. These synthetic dimensionless values illustrate a constraint error, not market savings. Correcting the author’s earlier expression and interpretation is a prerequisite for consistency, not the claimed contribution of this note.

The accompanying discrete OU verification permits unrestricted controls, uses the exact sampled OU transition, and charges a specified discrete interval exposure. That interval cost is a discrete-model assumption rather than an exact integral of every continuous-time cost component. Numerical checks of that model and of (2) support the algebra within their stated assumptions. They neither identify the empirical BTC process as OU nor validate the hypothetical replay costs. In particular, an unbounded Gaussian adaptive signal does not generally admit a nonzero fixed exposure cap guaranteeing monotone trades on every sample path.

5 Related literature and the scope of the contribution

This supplement distinguishes the paper’s empirical question from established results in predictive execution, uncertainty-aware trading and decision-focused learning. The study follows the author’s execution preprint [Lai, 2026], but its contribution must be assessed through the additional forecast comparison, chronological experiments and reproducible diagnostics. Neither the cryptocurrency setting nor the use of signals in execution is itself new. The relevant question is whether a forecast that predicts a terminal return also identifies price differences across the times at which a fixed order can actually be executed.

5.1 Predictive execution and the temporal shape of forecasts

The impact–risk trade-off in liquidation is foundational [Almgren and Chriss, 2001]. Subsequent work incorporates information about order flow and future prices into trading controls. Cartea and Jaimungal [2016] show how weighted expected future net order flow changes an execution strategy. Bechler and Ludkovski [2015] incorporate dynamic order-flow imbalance and consider an endogenous execution horizon, while Cartea et al. [2018] investigate trading decisions informed by order-book signals. These approaches already make the timing of information economically relevant. They cannot be represented adequately by a general claim that earlier execution research ignores predictive information.

Several papers are especially close to the temporal-profile argument. The deterministic, time-zero-conditioned formulation in Lehalle and Neuman [2019] uses the integrated expected signal across execution times and removes a fixed initial-price term from the objective. Forde et al. [2022] formulate execution with Gaussian signals and power-law resilience using a conditional remaining price change. The general-propagator formulation of Abi Jaber and Neuman [2025] likewise contains conditional price contrasts relative to the terminal price. Thus, the fact that relative expected prices matter for allocating a fixed quantity across time is already embedded in established execution objectives. The centered-profile identity used here is an elementary finite-grid representation of that fact, rather than a new principle of optimal execution.

The distinction is also related to the persistence of predictors. In dynamic portfolio choice with transaction costs, [Gârleanu and Pedersen \[2013\]](#) derive adjustment toward an aim portfolio that depends on future expected targets. That setting differs from compulsory fixed-volume liquidation, but it establishes why a predictor’s temporal behavior can affect its trading value. Learning latent alpha states [[Casgrain and Jaimungal, 2019](#)], adapting execution to signals under temporary and transient impact [[Neuman and Voß, 2022](#)], and responding to imminent trade signals under liquidity-dependent impact [[Bank et al., 2026](#)] provide further examples of richer information structures than a single terminal-direction forecast.

Our narrower diagnostic holds the feasible set and other cost terms fixed. A common additive shift in all execution-slot price forecasts contributes the same amount to every equal-volume schedule. It therefore cannot alter their linear price comparison. A constant drift is different: it generally changes expected prices unevenly across execution times. Nor does level cancellation establish invariance when forecasts change constraints, impact parameters or the normalization of a reported performance ratio. The empirical comparison concerns the centered temporal profile under the stated replay assumptions; it does not establish a general dominance result over these predictive-control models.

5.2 Uncertainty, shrinkage and fallback policies

Uncertainty-aware trading is also an established direction. [Cartea et al. \[2017a\]](#) study model uncertainty in algorithmic trading, primarily in a market-making setting. Liquidation under ambiguity aversion is addressed directly in [Cartea et al. \[2017b\]](#), and [Horst et al. \[2022\]](#) study liquidation under factor uncertainty. [De March and Lehalle \[2018\]](#) examine uncertainty associated with threshold-based signal controls. These are distinct formulations, rather than interchangeable guarantees for a fitted scalar attenuation rule. Portfolio parameter uncertainty [[Kan and Zhou, 2007](#)] supplies a related motivation for cautious estimation, but its portfolio-choice conclusions do not automatically transfer to an execution schedule.

The closest recent uncertainty-gating comparison is [Irshad and Biswas \[2026\]](#). Their execution framework gates next-interval forecast-driven action using uncertainty estimates and otherwise uses a VWAP fallback. Consequently, ignoring an uncertain forecast or retaining a baseline is not the novelty claimed here. Their equity-market experiment also cannot validate a cryptocurrency replay. The relevant differences are the quantity being diagnosed, the temporal forecast representation, the calibration procedure and the evidence produced under our own implementation.

Errors in execution-cost models require separate attention. [Hey et al. \[2023\]](#) analyze costs of misspecified price impact, while [Neuman et al. \[2023\]](#) propose pessimistic offline learning for propagator models. Stochastic impact enters the control problem in [Fouque et al. \[2022\]](#); [Macri and Lillo \[2024\]](#) investigate reinforcement learning when liquidity varies over time. These studies motivate treating a fixed impact coefficient as an assumption rather than a measured market property. Baseline-constrained policy improvement is also established in reinforcement learning [[Laroche et al., 2019](#)], but its safety results do not constitute guarantees for our calibration or resampling procedure. Here, attenuation and fallback are evaluated empirically within the specified replay.

5.3 Prediction quality, decision quality and coverage

The difference between statistical prediction accuracy and downstream decision quality is central to smart predict-then-optimize learning [[Elmachtoub and Grigas, 2022](#)]. [Liu and Grigas \[2021\]](#) establish surrogate-risk and calibration results under particular assumptions on the optimization

problem. The survey and benchmark of Mandi et al. [2024] place such methods within the broader decision-focused learning literature, and Gupta and Huang [2024] develop a directional-gradient approach. These works precede the motivation for evaluating forecasts through their consequences for a constrained decision. The present diagnostic is task-specific and does not introduce a general decision-focused learning framework or inherit its theoretical guarantees.

Temporal dependence adds a further distinction. Liu et al. [2024] investigate predict-then-optimize risk bounds with dependent data, illustrating why independence-based conclusions require additional work in time series. Prediction-interval coverage is likewise different from an execution-cost guarantee. Adaptive conformal inference targets coverage under distribution shift [Gibbs and Candès, 2021]; departures from exchangeability are explicitly analyzed by Barber et al. [2023]. Time-series adaptations include Zaffran et al. [2022] and Xu and Xie [2021], with different constructions and conditions. These references provide context for interpreting uncertainty-aware execution methods. They do not justify calling empirical shrinkage, a bootstrap interval or a baseline fallback distribution-free protection against trading losses.

5.4 Cryptocurrency evidence and backtest interpretation

Cryptocurrency execution has direct precedents. Bundi et al. [2024] study optimal trade execution in cryptocurrency markets, and Schnaubelt [2022] investigate cryptocurrency limit-order placement using reinforcement learning and detailed market data. The latter problem includes information about orders and trades that a bar-price replay cannot reproduce. Barucci et al. [2023] examine market impact and efficiency in cryptoassets, while Makarov and Schoar [2020] document exchange segmentation and arbitrage-related price differences. The microstructure review of Almeida and Gonçalves [2024] supplies broader context. Together these studies motivate asset-, exchange-, horizon- and data-resolution-specific interpretation; they do not establish profitable execution from terminal-return predictability alone.

Finally, dependence and model selection are separate sources of uncertainty. The stationary bootstrap of Politis and Romano [1994] is a specific dependence-aware resampling construction, not a blanket justification for every block-bootstrap implementation. Bailey et al. [2017] examine overfitting created by selecting among backtests, and Arnott et al. [2019] advocate a disciplined empirical research protocol. Accordingly, chronological splits, calibration choices, inspected periods and subsequent design changes should be disclosed. An expanded study designed after an initial result has been observed remains exploratory in that respect. The contribution is a transparent comparison of forecast targets and execution consequences under stated assumptions, with uncertainty and negative findings reported alongside any apparent gains.

6 Availability and remaining submission steps

The package contains the original frozen study, the separately versioned extension protocol and code, sixteen archived inputs and checksums, complete machine-readable results, verification results, and manuscript sources. The submission snapshot is available at <https://github.com/JienWeng/crypto-execution-timing/tree/v1.0.0>. The data manifest’s statement that no outcomes had been computed describes its creation stage, not the final state of the project.

OpenAI Codex assisted source discovery, mathematical and code analysis, drafting and internal review. The internal reviews used the same model family; they are not external journal peer review or a substitute for author responsibility. The author has approved the manuscript, authorship and the stated Monash University affiliation. No funding was received for this research. The author declares no competing interests. The confirmed correspondence address is

lai.jienweng@monash.edu. The work uses public aggregate market data, without human participants or live orders; no institutional ethics determination is asserted.

Applied Economics Letters is the author-confirmed target. The current official instructions were inaccessible during the final literature check, so exact word-count inclusions and supplementary-material acceptance remain unverified. The concise article and separate supporting review are prepared for adaptation once those requirements are checked. The broad library contains forty research papers, including explicitly identified preprints and reviews, plus a data-documentation record; twenty-five papers date from 2020–2026. This is a targeted literature synthesis rather than a systematic search or proof of novelty.

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